FINDINGS REPORT

1. Feature Importance Analysis

Using a Random Forest Regressor, feature importances were extracted to identify primary price drivers:

- **Dominant Feature:** Living area square footage (sqft_living) accounts for **97.85% of price variance.**
- **Secondary Features:** Waterfront presence (0.60%), grade (0.53%), and bedrooms (0.29%) provide minor marginal predictive value.

2. Model Performance Leaderboard

Evaluated on an 80/20 train-test split ($N = 200$ test samples):

Model	MAE ($)	RMSE ($)	R2 Score
Lasso Regression ($\alpha = 1000.0$)	**15,363.22**	**19,157.01**	**0.9966**
Linear Regression	15,416.42	19,198.67	0.9966
Ridge Regression ($\alpha = 1.0$)	15,434.61	19,199.19	0.9966
Gradient Boosting Regressor	21,650.81	27,551.60	0.9929
Random Forest Regressor	28,768.15	36,905.59	0.9873

Sample Prediction Usage

```
Python
import pickle
import pandas as pd

# Load model and scaler
with open('models/house_price_model.pkl', 'rb') as f:
    model = pickle.load(f)

with open('models/scaler.pkl', 'rb') as f:
    scaler = pickle.load(f)

# Sample property input
new_house = pd.DataFrame([{
    'bedrooms': 3, 'bathrooms': 2, 'sqft_living': 2800, 'sqft_lot': 5000,
    'floors': 2, 'waterfront': 0, 'view': 1, 'condition': 3, 'grade': 7,
    'year_built': 2005, 'year_renovated': 0, 'lat': 47.55, 'long': -122.10,
    'house_age': 21, 'is_renovated': 0, 'renovation_age': 0
}])

scaled_input = scaler.transform(new_house)
predicted_price = model.predict(scaled_input)[0]

print(f"Estimated Market Value: ${predicted_price:,.2f}")
```